

How China's Policies about 'People' Drive AI Development

The Synergistic Effect of Policy, Talent, and Industry

Group Eight

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Abstract

Abstract: This research investigates the mechanisms through which China's "people-centric" policies have propelled its rapid advancement in Artificial Intelligence (AI). The study focuses on the synergistic interplay between government initiatives and talent development, the strategic utilization of widespread public participation and data contribution, and the collaborative efforts of policy, academia, and industry. Through an analysis of national AI strategies, talent cultivation programs, data governance frameworks, and case studies of leading AI enterprises, this research finds that human capital—encompassing elite researchers, skilled developers, and the general populace—plays a critical role in fostering innovation and shaping China's AI trajectory.

Keywords: AI Development; China; Human-Centric Policies; Talent Development; Data Governance

Introduction

Research Background, Objectives & Significance

Research Background

The Rise of China's AI Industry

- China has rapidly emerged as a **formidable power** in the global AI landscape
 - Significant technological achievements like **DeepSeek-V3 model**
 - AI industry estimated worth **nearly 600 billion RMB** by late 2024
 - Projected for continued significant expansion
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Government Strategy

- **"Next-Generation AI Development Plan"** (2017): Goal to become world leader by 2030
- Target: Core AI industry scale to exceed **1 trillion RMB** by 2030
- Emphasis on strategic integration of **"talent, technology, and industry chains"**
- Focus on leveraging human value across entire spectrum

Research Objectives

Core Research Question

What are the primary drivers behind China's rapid AI advancement, focusing on people-centric policies that integrate talent development and public participation?

Specific Sub-Questions

1. How do Chinese government policies effectively **empower AI talent**?
2. How does widespread **public engagement** accelerate AI innovation?
3. What is the nature of **synergy** between government, academia, and industry?
4. What are the **measurable outcomes** of people-centric strategies?
5. What are the key **challenges and limitations** of this model?

Research Significance

Theoretical Significance

- Provides comprehensive explanation for China's **rapid growth trajectory**
- Moves beyond simplistic narratives of state funding
- Uncovers nuanced interplay of **human capital development**
- Contributes to comparative AI governance discourse

Practical Significance

- Offers lessons for other nations fostering **AI ecosystems**
- Informs global strategies for **cultivating AI talent**
- Guides **ethical data resource management**
- Promotes inclusive and sustainable **AI development**

Literature Review

Current Research on AI Policy & Talent Development

Current Research on AI Policy & Talent Development

The literature highlights contrasting AI development paths between China and the EU. China is transforming into an innovation-driven economy with substantial AI investments through multi-channel funding and talent initiatives. Despite efforts to attract international researchers, China faces cultural and institutional barriers limiting its effectiveness compared to competitors.

The EU leads in AI ethics discussions but lacks sufficient funding and policy support for technological development. Critical challenges include absence of indigenous tech giants and data monopolization by non-EU firms.

Research reveals a gap in understanding the mechanisms through which people-centric policies drive AI advancement, particularly the synergistic relationship between government policy, talent development, and public participation that characterizes China's approach.

Research Methods

Mixed-Methods Approach

Research Methods Overview

Mixed-Methods Approach

Qualitative Methods:

- **Policy Document Analysis:** 20+ core documents (2020-2024)
 - **Industry Case Studies:** 5 leading Chinese AI companies
 - **Thematic Analysis:** Identifying strategic priorities
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Quantitative Methods:

- **Statistical Data Analysis:** AI industry metrics
- **Patent Filing Analysis:** Trends and quality assessment
- **Investment Data:** Public and private funding

Data Collection & Analysis Framework

Data Sources

- **Government Documents:** State Council, MIIT, Ministry of Education
 - **Industry Reports:** CAICT, iResearch, World Economic Forum
 - **Corporate Data:** Company reports, disclosures, case studies
 - **Academic Publications:** Peer-reviewed research
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Analysis Methods

- **Correlation Analysis:** Policy implementation vs. industry growth
- **Comparative Analysis:** China vs. US and EU approaches
- **Triangulation:** Multiple data sources for validation
- **Time-Series Analysis:** Tracking development over time

Research Results

The Four Pillars of China's People-Centric AI Model

The Four Pillars Overview

Policy

The Governmental Blueprint

- National AI strategies
- Talent cultivation policies
- Data governance frameworks

Talent

Cultivating Human Intelligence

- University programs expansion
- Industry training initiatives
- International talent attraction

Public Participation

Masses as Data Contributors

- Vast user data generation
- AI application feedback
- Model training resources

Industry

Engine of Innovation

- Leading tech companies (BAT)
- Open source initiatives
- Industry-academia collaboration

Pillar 1: Policy - The Governmental Blueprint

Pillar 2: Talent - Cultivating Human Intelligence

Talent Landscape & Statistics

- **47%** of world's top-tier AI researchers produced by China
- **535 universities** offering AI-related majors
- **4 million** projected AI talent gap by 2030
- Above global average growth in AI/ML specialists

University Initiatives

- **50 AI research centers** established by 2020
- Curriculum integration across disciplines
- "New engineering" and "new science" programs

Industry Training Programs

Baidu: Plan to cultivate **10 million AI experts** over 5 years, collaboration with **700+ universities**

Huawei: "**3-in-1**" **approach** with universities, **ICT Competition** for talent development

Tencent: "**Qingyun Plan**" - 100+ top technical talents annually, comprehensive career stage training

Pillar 3: Public Participation - Masses as Data Contributors

Scale of User Data

- **Over 1 billion** internet users in China
- Massive data from social media, e-commerce, payments
- Strategic asset for AI model training

Ethical Governance

- **PIPL** ensures consent and user rights
- **AIGC measures** require content labeling
- Balance between innovation and protection

Data-Driven Improvements

The continuous interaction between users and AI applications creates a **dynamic data loop** for innovation acceleration.

Major Platform Data Utilization

Baidu: 704M MAUs for Baidu App, search query data for ERNIE models

iFLYTEK: 800M users, voice recognition accuracy improved from **60% to 98%**

Alibaba: E-commerce data for recommendation engines, **35% conversion rate increase**

Tencent: Social interaction data from WeChat/QQ for personalization

Pillar 4: Industry - Engine of Innovation

Industry Growth

- **600 billion RMB** core AI industry scale (2024)
- **4,500+ related companies**
- Projected **1.73 trillion RMB** by 2035
- **15.6% CAGR** (2025-2035)

Investment Trends

- Private AI investments: \$650M (2023) → \$3.15B (2024)
- Government participation in 60%+ of generative AI deals
- Foreign investment beginning to show interest

Open Source Movement

PaddlePaddle (Baidu):

- China's first independently developed framework
- 18.1M developers community

OpenI Community:

- Open AI collaboration platform
- DeepSeek's open-source models

Industry Adoption:

- 100+ mainstream models supported
- Global AI commons contribution

Synergistic Effects

Government-Academia-Industry Collaboration

National Labs:

- **Shanghai AI Laboratory (SHLAB):** 13 university collaborations
 - **Beijing Academy of AI (BAAI):** WuDao model with 1.75T parameters
 - **Joint Research Labs:** Huawei-university partnerships
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Feedback Loops

- **Data-Driven Policy:** User adoption rates inform policy adjustments
 - **Industry Innovation:** Real-world user feedback drives product development
 - **Talent Flow:** Movement between academia and industry
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National Champions Strategy

- Large companies (Baidu, Alibaba, Tencent, Huawei) as talent magnets
- Platform creation for smaller developers
- Technical standards leadership

Discussion

Key Findings & Contributions

Key Findings

Success Formula Components

1. Strategic Human Capital Management

- Systematic integration of elite talent and public participation
 - Multi-level talent cultivation from universities to industry
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2. Synergistic Policy Framework

- Coordinated government-academia-industry collaboration
 - Balance between innovation promotion and ethical governance
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3. Scale Advantage

- Massive user base providing unprecedented data resources
- Large domestic market for AI application testing
- Rapid iteration and improvement cycles

The true potency lies in the **synergistic effects** generated by the interaction of all four pillars working together.

Theoretical Contributions

Academic Contributions

- **Conceptual Framework:** Four-pillar model of AI development
- **Mechanism Identification:** How human capital drives AI advancement
- **Comparative Perspective:** China's model vs. Western approaches
- **Policy Analysis:** Evidence-based evaluation of effectiveness

Practical Implications

- **Policy Guidance:** Lessons for other nations
- **Talent Strategy:** Best practices in AI workforce development
- **Data Governance:** Balancing innovation and protection
- **Industry Development:** Ecosystem building approaches

Conclusion

Research Summary & Future Directions

Research Summary

Core Conclusion

China's AI development success stems from a **systematic and strategic focus on 'people'** through:

1. **Policy Architecture:** Comprehensive frameworks prioritizing human capital
2. **Talent Engine:** Multi-channel cultivation and attraction mechanisms
3. **Public Participation:** Leveraging mass user data and engagement
4. **Industry Innovation:** Dynamic private sector with strong talent focus
5. **Synergistic Effects:** Coordinated interaction of all pillars

The "People's Algorithm"

Where human intellect and human-generated data are systematically integrated and amplified by policy and industry to create a powerful engine for AI innovation.

Limitations & Future Work

Research Limitations

- **Methodological:** Policy interpretation bias, data access constraints
- **Scope:** Focus on 'people' may underemphasize other factors
- **Temporal:** Rapidly evolving landscape, snapshot in time
- **Generalizability:** China-specific context may limit transferability

Future Research Directions

- **Longitudinal Studies:** Track long-term impacts
- **Comparative Analysis:** Detailed cross-country comparisons
- **Ethical Implications:** Deeper exploration of data governance
- **Implementation Studies:** How other nations adapt these lessons
- **Technology Evolution:** Impact of emerging AI technologies

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Thank You

Questions & Discussion

The People's Algorithm: How China's Human-Centric Policies Drive AI Development